MAT454

Made with Goodnotes

$\mathbb{P}^1 = \mathbb{C}^2 \setminus 0 / \mathbb{C}^\times$. $\quad \mathcal{O}(-1) = \{([z_0:z_1], \lambda(z_0,z_1)) \in \mathbb{P}^1 \times \mathbb{C}^2 : \lambda \in \mathbb{C}\}$.

$$\begin{tikzcd} \mathcal{O}(-1) \arrow[d, "\pi"] \\ \mathbb{P}^1 \end{tikzcd} \qquad \begin{tikzcd} ([z_0:z_1], \lambda(z_0,z_1)) \arrow[d, mapsto] \\ {[z_0:z_1]} \end{tikzcd}$$

$U_i = \{[z_0:z_1] \in \mathbb{P}^1 : z_i \neq 0\}$ $\quad i = 0,1.$

$$\begin{tikzcd} U_i \arrow[d, "\varphi_i"'] \\ \mathbb{C} \end{tikzcd} \qquad [z_0:z_1] \longmapsto \frac{z_{1-i}}{z_i} =: w_i$$

1. Show $\sigma_0(w_0) = (1, w_0)$ is section of $\mathcal{O}(-1)$ over U_0.
   $\sigma_1(w_1) = (w_1, 1)$ section over U_1.

$$\begin{tikzcd} w_0 \in \mathbb{C} \arrow[r, mapsto, bend left, "{[1,w_0] \mapsto ([1,w_0], (1,w_0))}"] & \mathcal{O}(-1) \arrow[d, "\pi"] \\ & U_0 \subset \mathbb{P}^1 \arrow[d, "{[z_0,z_1] \mapsto \frac{z_1}{z_0} =: w_0}"] \\ w \mapsto [1,w] \arrow[r, mapsto, bend left] & \mathbb{C} \end{tikzcd}$$

$\sigma_0 : U_0 \longrightarrow \mathcal{O}(-1)$
$\quad [z_0:z_1] \longmapsto ([z_0:z_1], \frac{1}{z_0}(z_0,z_1))$

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$\psi_0 : U_0 \times \mathbb{C} \xrightarrow{\sim} \mathcal{O}(-1)|_{U_0}$
$\quad ([z_0:z_1], \lambda) \longmapsto ([z_0:z_1], \frac{\lambda}{z_0}(z_0,z_1))$

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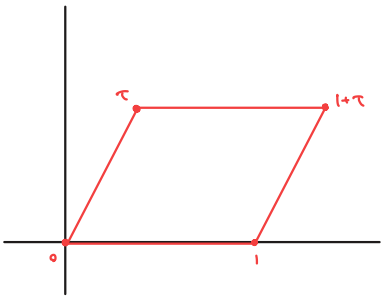
$\quad ([z_0:z_1], \lambda) \longmapsto ([z_0:z_1], \frac{\lambda}{z_1}(z_0,z_1))$.

2. $\exists\, \rho_{01}$ nonvanishing hol. fn st. $\sigma_0 = \rho_{01}\sigma_1$
   $\rho_{01} = \psi_0 \circ \psi_1^{-1}$.

3. $D \subset U_{01}$, $\psi : D \longrightarrow \mathcal{O}(-1)|_D$ loc hol sectn.
   $= f_0\sigma_0$
   $= f_1\sigma_1$

   Show $f_0 = \underbrace{t_{01}}_{= \rho_{01}^{-1}} f_1$.

4. $\mathcal{O}(-1)$ has metric $\langle ([z_0:z_1], \lambda(z_0,z_1)), ([u_0:u_1], \mu(u_0,u_1)) \rangle := \langle \lambda(z_0,z_1), \mu(u_0,u_1) \rangle$
   $= \lambda\bar{\mu}(u_0z_0 + u_1z_1)$.



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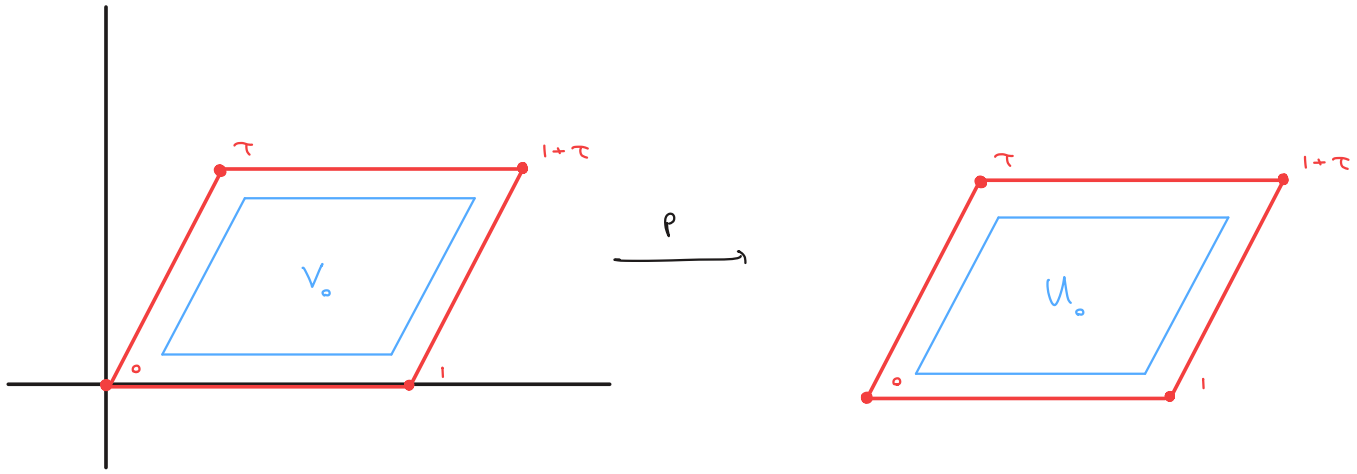

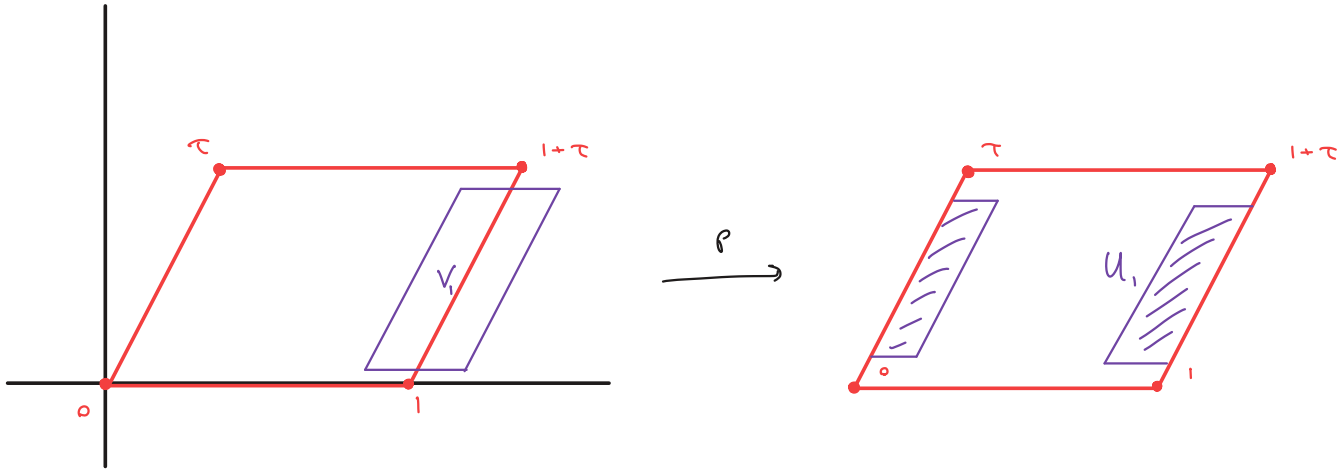

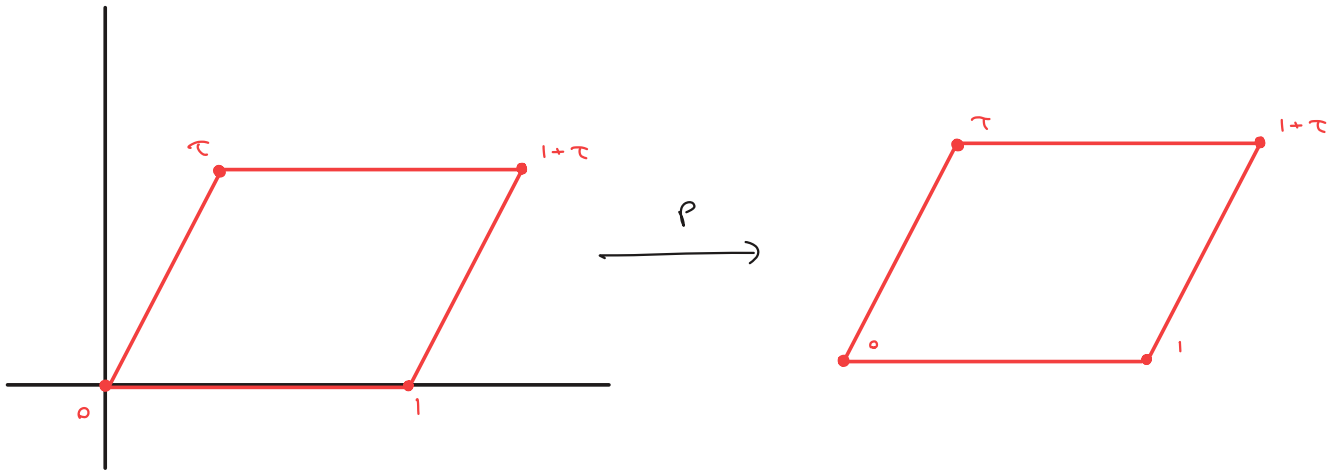

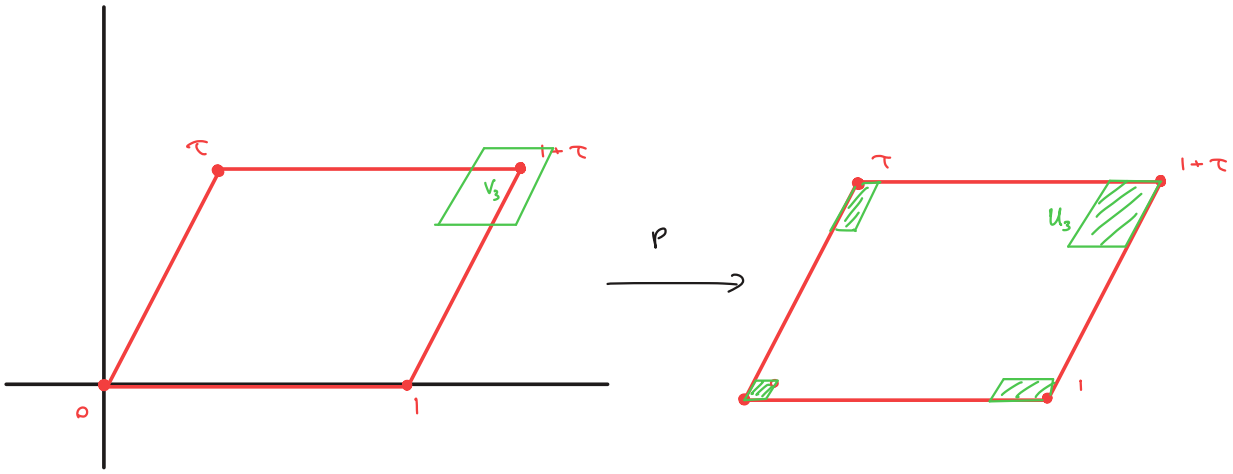

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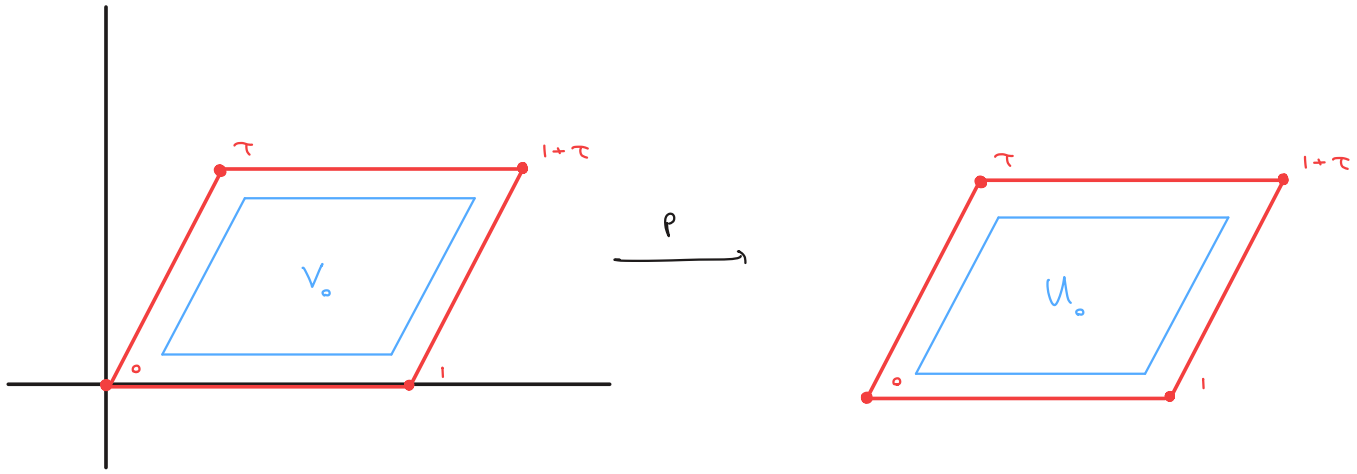

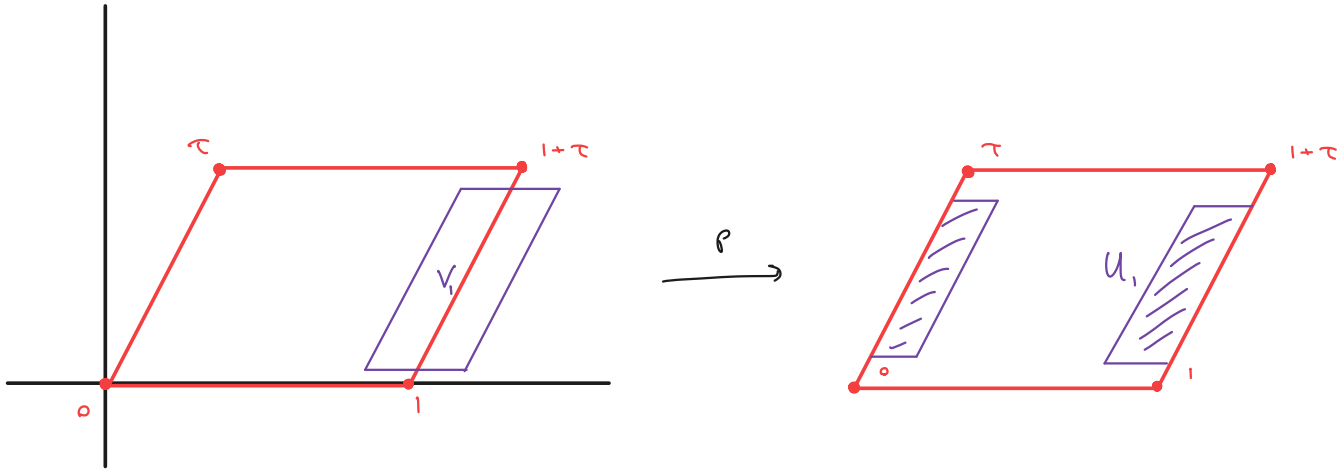

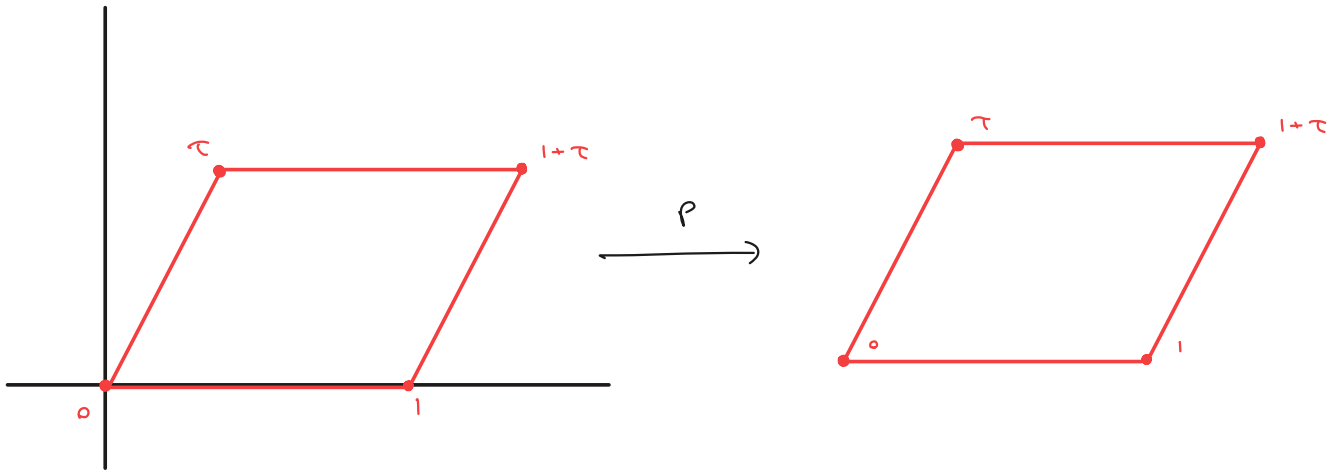

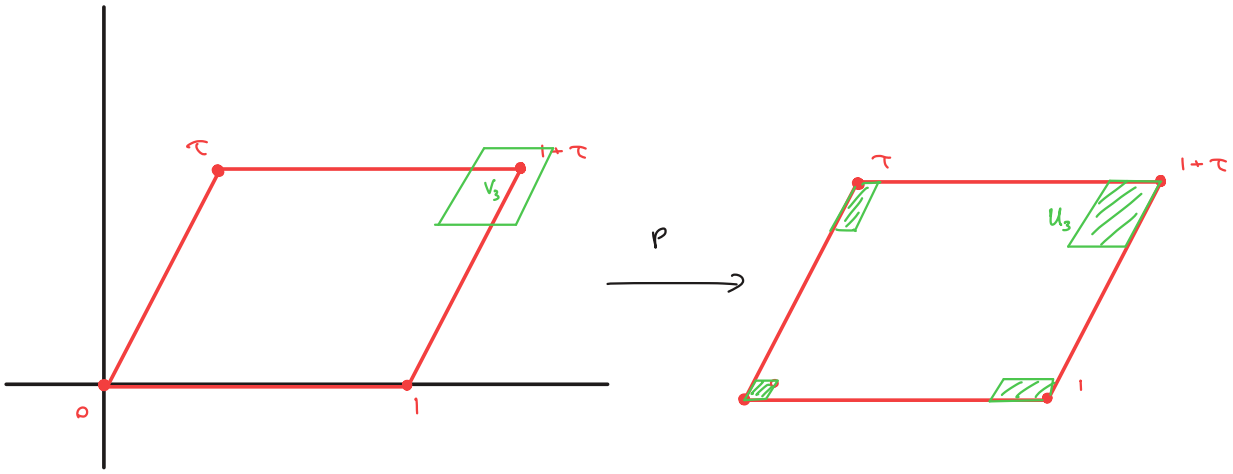

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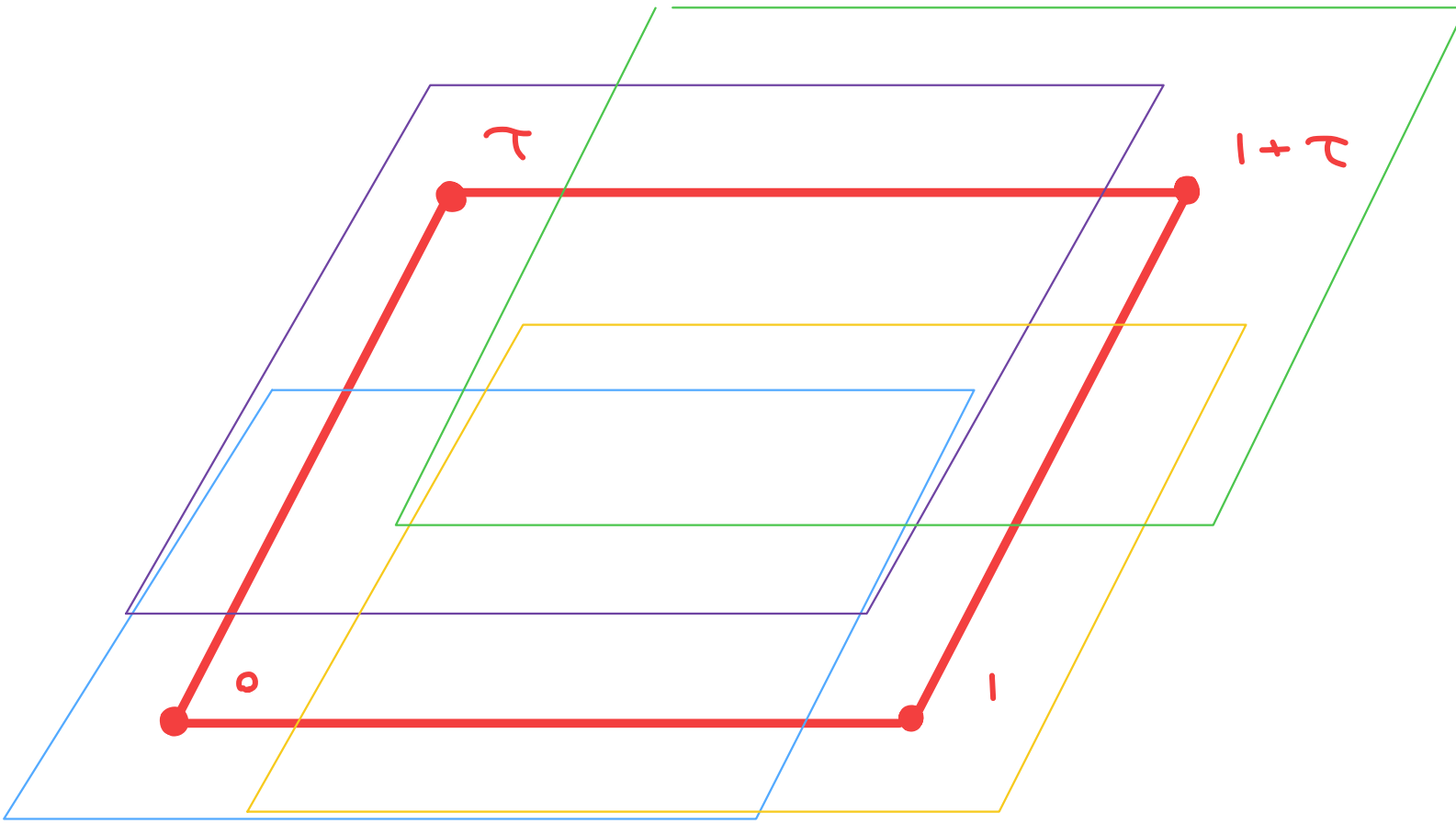

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$$\theta_1(z+1) = -\theta_1(z)$$
$$\theta_1(z+\tau) = -e^{-\pi i\tau - 2\pi i z}\,\theta_1(z).$$

$$\begin{array}{c} \mathcal{L} \\ \pi \downarrow \\ X \end{array}, \quad \text{trivialization } \mathcal{U} = \{U_i\} \text{ st. } \begin{array}{ccc} U_i & \subset & \mathcal{L} \\ \varphi_i \downarrow & \nearrow \theta_1 & \\ \mathbb{C} & & \end{array}.$$

i.e. if z_i is coord on U_i, z_j U_j

then $\theta_1(z_i) = t_{ij}(z_j)\,\theta(z_j)$

$t_{ij}: U_{ij} \longrightarrow \mathbb{C}^\times$. hol

$D \geq 0$ if $\alpha(P) \geq 0$ $\forall P$

>0 if also $\alpha(P) > 0$ $\exists P$.

$D_1 \geq D_2$ (resp $>$) if $D_1 - D_2 \geq 0$ ($>$)

$0 \neq f \in \mathcal{K}(M)$ is multiple of D if $(f) \geq D$

where $(0) \geq D$ $\forall D$.

thus f is multiple if $f = 0$ or $\mathrm{ord}_P f \geq \alpha(P)$ $\forall P \in M$

- if $\alpha(P) = 0$, f hol.
- if $\alpha(P) > 0$, f has zero ord $\geq \alpha(P)$
- if $\alpha(P) < 0$, f may have pole ord $\leq -\alpha(P)$.

$L(D) := \{f \in \mathcal{K}(M) : (f) \geq D\}$ mero fns w zeroes/poles no worse than D. if $\mathrm{ord}_P(0) = 0$, f hol, ... e.g $L(0) = H^0(M) = \mathbb{C}$.

$r(D) := \dim L(D)$. dimension

$\Omega(D) := \{\omega$ mero diff : $(\omega) \geq D\}$ mero diff no worse than D. eg. $\Omega(0) = \mathcal{H}^1(M)$.

$i(D) = \dim \Omega(D)$ index of speciality

r, i depend only on divisor class. if $0 \neq \omega$, $i(D) = r(D(\omega)^{-1})$.

Thm (R-R) $r(-D) = \deg D - g + 1 + i(D)$

$\dim H^0(X, \mathcal{O}(D))$ $\qquad$ $\dim H^0(X, \Omega(-D))$

$$h^0(X, \mathcal{O}_X(D)) - h^0(X, K_X(-D)) = \deg D + 1 - g$$

$\dim\{f$ mero : $f = 0$ or $(f) \geq -D\}$ $\qquad$ $\dim\{\omega$ mero : $\omega = 0$ or $(\omega) \geq D\}$

$$\ell(D) - \ell(K-D) = \deg D + 1 - g.$$

$\dim H^0(X, \mathcal{O}(D))$ $\qquad$ $\dim H^0(X, \Omega_X(-D))$

$h^0(X, \mathcal{O}_X(-D))$?

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